What Survives Diffusion Noise? Exact Scores, Order-by-Order Information Decay, and Transition Learning in Non-Gaussian Markov Chains

Anonymous Authors[1]

Abstract

Diffusion models are trained to approximate the score of a noised data distribution, a target that is unknown for almost every distribution of interest. We study a family in which it is known exactly and non-trivial: a first-order Markov chain with continuous, non-Gaussian transitions under coordinatewise Ornstein–Uhlenbeck noise, where conditioning leaves the posterior factor graph a tree and one forward–backward sweep of functional message passing returns the exact score. This laboratory lets us separate what diffusion noise does to different orders of structure. We prove that the channel attenuates every Hermite mode of degree m by e^{-mt}, so that the Fisher information one noised sequence carries about an order-m feature decays as e^{-2mt}: correlation (degree two) survives as e^{-4t}, non-Gaussian shape (degree four) as e^{-8t}, and the gap between the true score and that of the covariance-matched Gaussian—which we show is exactly what Gaussian message closure computes—is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, with the crossover set by the covariance spectrum. Tree-exact inference verifies the rates and the constants. We then learn the transition law from noised sequences alone, by expectation–maximisation around the same recursion with Fisher's identity supplying the exact gradient, and show that it recovers fourth-order structure from strictly less information than the denoisers it is compared against: one noised view per sequence, never a clean chain, against clean targets with a fresh corruption at every gradient step. [PENDING: final baseline, generation and misspecification numbers; the abstract is rewritten once §5 is final]

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Generative diffusion produces samples by reversing a noising process, and the reversal is driven by the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ of the noised data law (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021). Training regresses a network onto that score through denoising (Vincent, 2011), and every question about what such a network has learned is a question about the distance between what it computes and the score it should have computed. For realistic data the target is unavailable, so a large error cannot be separated from a hard target; the tractable alternatives are Gaussian, whose scores are linear, or mixtures, whose nonlinearity lives in the mixture index rather than in any dependence between coordinates.

We study a family that is neither. A first-order Markov chain with continuous, non-Gaussian transitions has a globally dependent, nonlinear score, yet under coordinatewise Ornstein–Uhlenbeck noise its posterior factor graph is a tree, so the score is computable exactly by two sweeps of functional message passing (Section 2). Because the chain's covariance is fixed by two numbers and its higher-order structure by the innovation law, the model separates second-order from higher-order structure by construction: Gaussian, Laplace, Student-t, uniform and bimodal innovations, and a one-parameter family that interpolates between them, share one covariance exactly.

The question. What information about structured data survives diffusion noise, and how much does knowing the structure reduce the data needed to recover the score and the dynamics that generate it?

Diffusion noise is a low-pass filter on statistical order. The first answer is a theorem (Section 3). Under the

Ornstein–Uhlenbeck channel every Hermite mode of degree m is attenuated exactly by e^{-mt}, so the Fisher information a noised sequence carries about an order-m feature of the data decays as e^{-2mt}. Correlation lives at degree two and decays as e^{-4t}; non-Gaussian shape first appears at degree four and decays as e^{-8t}; and the gap between the true score and the score of the covariance-matched Gaussian—which we prove is exactly what single-Gaussian message closure computes (Proposition 1)—is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ to leading order, with an explicit crossover $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ before which the rates are shallower. The exact score lets us verify the exponents and the constants (Figure 2), and the result explains, rather than contradicts, earlier measurements on this model that stopped before t_*.

Learning through the channel. The second answer is empirical. We learn the transition law from noised sequences alone, by expectation–maximisation around the same exact recursion, with Fisher's identity supplying the exact gradient of the marginal likelihood from one forward–backward pass (Section 4). The theorem predicts that the shape of the innovation law is recovered e^{4t} times more slowly than its correlation at observation level t; the experiment confirms it, and locates the levels at which shape is still learnable. Against denoising networks trained on the clean trajectories—a fully connected network, a weight-shared window head, and dilated, recurrent, attention-based and multiscale architectures under one training recipe—the structured estimator is ahead at every sample size while seeing strictly less: one noised view per sequence, never a clean chain, against clean targets with a fresh corruption at every gradient step (Section 5). Given enough clean data the networks do cross the second-order floor of Proposition 1, so the gap is a statement about the information interface and not about a barrier they cannot pass. We show that this advantage is earned by the non-Gaussian capacity of the innovation model and not only by the Markov factorisation, that it transfers to the generated distribution, that it extrapolates in sequence length, and that it has a measurable boundary under misspecification.

Contributions. (i) An exactly solvable non-Gaussian, correlated diffusion laboratory with a validated score reference, and the identification of Gaussian message closure with the LMMSE of the covariance twin (Proposition 1). (ii) Theorems 2 and 3: exact Hermite attenuation and the order-by-order decay of score gaps and Fisher information, with constants and crossover, verified with the exact score. (iii) Transition learning from noised data by exact-inference EM, with the predicted e^{4t} separation of shape from correlation recovery measured. (iv) A controlled comparison against learned denoisers—capacity attribution, fair architecture screen, information-matched training, generation, length extrapolation and misspecification—showing what explicit structure buys and where it stops.

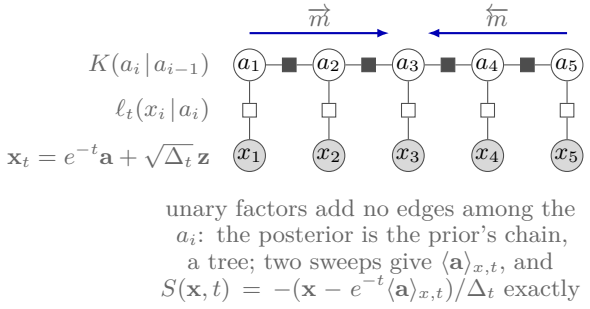


Figure 1. Why the score is exact. Coordinatewise noising attaches one unary factor per site to the chain prior and adds no edge among the latent variables, so the posterior factor graph is the prior's chain; two message sweeps return the posterior mean, and Tweedie's identity turns it into the score.

2. Setup: an exactly solvable non-Gaussian laboratory

Data and channel. A clean sequence $\mathbf{a} \in \mathbb{R}^L$ is a first-order Markov chain,

$$P_0(\mathbf{a}) = \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1}), \quad K(a' \mid a) = \varphi(a' - \rho a), \tag{1}$$

with $a_1 \sim \mathcal{N}(0,1)$, $|\rho| < 1$, and an innovation density φ of mean zero and variance $q = 1 - \rho^2$. The variance choice makes $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly for every innovation law, so the families we use—Gaussian, Laplace, Student-t, uniform, a symmetric two-component mixture, and the generalised Gaussian $\varphi_\beta(\varepsilon) \propto \exp(-|\varepsilon/s_\beta|^\beta)$ whose exponent β moves the excess kurtosis from $+3$ (Laplace) through 0 (Gaussian) to -1.1 at fixed variance—share one covariance $\mathbf{\Sigma}_0$ and differ only through cumulants of order four and higher. The forward process is the coordinatewise Ornstein–Uhlenbeck channel

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}), \tag{2}$$

so $\mathrm{Var}(x_{t,i}) = 1$ at every t and the noised law P_t tends to $\mathcal{N}(0, \mathbf{I})$.

The score is a posterior mean, and the posterior is a tree. Tweedie's identity gives $S(\mathbf{x},t) = -(\mathbf{x} - e^{-t}\langle\mathbf{a}\rangle_{x,t})/\Delta_t$, with $\langle\mathbf{a}\rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$ (Robbins,

1956; Efron, 2011). Conditioning Equation (1) on $\mathbf{x}$ multiplies the prior by one unary factor $\ell_t(x_i \mid a_i) \propto \exp[-(x_i - e^{-t}a_i)^2/2\Delta_t]$ per site: no edge is added among the latent variables, the posterior factor graph is the prior's chain, and sum-product on it is exact (Pearl, 1988; Kschischang et al., 2001). Two message sweeps,

$$\begin{aligned} \overrightarrow{m}_i(a) &\propto \int K(a \mid a')\, \ell_t(x_{i-1} \mid a')\, \overrightarrow{m}_{i-1}(a')\, \mathrm{d}a', \\ \overleftarrow{m}_i(a) &\propto \int K(a'' \mid a)\, \ell_t(x_{i+1} \mid a'')\, \overleftarrow{m}_{i+1}(a'')\, \mathrm{d}a'', \end{aligned} \tag{3}$$

return the site marginals $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ and hence the exact score of a correlated, non-Gaussian, L-dimensional law. Messages are functions on $\mathbb{R}$; we represent them on a grid of N_{g} points on $[-A, A]$ with trapezoidal quadrature, which makes the recursion exact for the discretised factor model and controlled for the continuous one (Appendix B: at $N_{\mathrm{g}} = 401$, $A = 8$ the reference is accurate to 10^{-5} relative on ordinary draws for every family used). Throughout, "the score" means this tree-exact functional inference, evaluated by validated grid quadrature.

The covariance twin and the price of second-order closure. The Gaussian chain with the same ρ and q has the closed-form score $S^{\mathrm{G}}(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$, $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$, and posterior mean $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$, the linear minimum-mean-square (LMMSE) estimator.

Proposition 1 (Gaussian closure computes the LMMSE of the covariance twin). Let Π_{G} project a density onto the Gaussian with its mean and variance, and let moment-projected sum-product apply Π_{G} to each message after every transition integral in Equation (3). For the chain Equation (1) with $\rho \neq 0$ and any innovation law of mean zero and variance q, it returns exactly the LMMSE estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$, the same for every such law.

The proof (Appendix A) is an induction over the two sweeps: a convolution's first two moments depend on the innovation only through its first two moments, so every projected message equals the Gaussian chain's message. Proposition 1 turns the gap between S and S^{G} into an interpretable object: it is precisely what any estimator that carries only second-order structure—a Gaussian innovation kernel, a linear denoiser, or the Kalman smoother—must leave on the table. Section 3 computes its size.

3. What survives diffusion noise: attenuation by statistical order

Write γ for the standard Gaussian on $\mathbb{R}^L$, He_k for the probabilists' Hermite polynomials and $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$ for their products, of degree $|\nu|$. The channel Equation (2) acts on Hermite moments diagonally.

Theorem 2 (Hermite attenuation). For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$, $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$.

Proof. Mehler's identity $\mathbb{E}_z[\mathrm{He}_n(\alpha a + \sqrt{1-\alpha^2}\, z)] = \alpha^n \mathrm{He}_n(a)$ for $z \sim \mathcal{N}(0,1)$ follows from the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$; coordinates are independent given $\mathbf{a}$, and $\alpha = e^{-t}$. □

Nothing here requires $P_0/\gamma \in L^2(\gamma)$—which fails for the Laplace chain and, because $\lambda_{\max}(\mathbf{\Sigma}_0) > 2$, for the Gaussian chain at $\rho = 0.85$—only finite moments. Two consequences are exact at every t: cumulants of order $m \geq 3$ scale as e^{-mt} (Gaussian noise has none), and the degree-4 Hermite moments of two variance-matched laws differ exactly by their fourth-cumulant tensors κ_4, attenuated by e^{-4t}. Correlation lives at degree two and decays as e^{-2t}; non-Gaussian shape first appears at degree four and decays as e^{-4t}. Diffusion noise is a low-pass filter on statistical order.

The consequence for scores and for learning is an expansion in the signal-to-noise ratio e^{-2t}. Let $s = e^{-t}\mathbf{x}/\Delta_t$, $b = e^{-2t}/\Delta_t$ and $\Lambda(s, b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$, so that $\log P_t(\mathbf{x}) = \log \gamma_{\Delta_t}(\mathbf{x}) + \Lambda(s, b)$ and $S = -\mathbf{x}/\Delta_t + (e^{-t}/\Delta_t)\nabla_s \Lambda$ exactly. Grading Λ by order in e^{-t}, its second-order part is the same for P_0 and its covariance twin, and its fourth-order part differs by a term linear in κ_4.

Theorem 3 (Order-by-order information decay). Let P_0 have finite eighth moments and covariance $\mathbf{\Sigma}_0$, let S^{G} be the score of its covariance twin, and let $\lambda_{\max} = \lambda_{\max}(\mathbf{\Sigma}_0)$. As $t \to \infty$,

$$\mathbb{E}_{P_t}\|S - S^{\mathrm{G}}\|^2 = \tfrac{1}{6}\|\kappa_4\|_F^2\, e^{-8t}(1 + \epsilon_t), \tag{4}$$

$$I_t(\theta) = \tfrac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2\, e^{-4t}(1 + \epsilon_t) \text{ if } \partial_\theta \mathbf{\Sigma}_0 \neq 0, \tag{5}$$

$$I_t(\theta) = \tfrac{1}{24}\|\partial_\theta \kappa_4\|_F^2\, e^{-8t}(1 + \epsilon_t) \text{ if } \partial_\theta \mathbf{\Sigma}_0 = 0, \tag{6}$$

the last with $\partial_\theta \kappa_4 \neq 0$, and with $\epsilon_t = O(e^{-2t}\lambda_{\max})$, where $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ is the Fisher information one noised sequence carries about a parameter θ of P_0, and $\|\cdot\|_F$ is the Frobenius norm of the matrix or symmetric 4-tensor. For a Gaussian chain Equation (5) is exact at every t with $I_t = \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_t^{-1}\partial_\theta \mathbf{\Sigma}_t \mathbf{\Sigma}_t^{-1}\partial_\theta \mathbf{\Sigma}_t)$.

The information statements Equations (5) and (6) are the diffusion-time form of a principle established for low-SNR latent-variable models: the log-likelihood expands in powers of the SNR with the order-k term carrying k-th-moment information (Katsevich & Bandeira, 2023; Perry et al., 2019; Bandeira et al., 2023), and Fisher information about directions first identifiable at moment order k scales as SNR^k (Balanov et al., 2026); with $\mathrm{SNR}_t \simeq e^{-2t}$ these are Equations (5) and (6). The cumulant attenuation behind Theorem 2 is also written down for the forward process by Aarts et al. (2025). What is new is the score-level statement Equation (4) with its constant, made interpretable by Proposition 1; the crossover below; and the exact verification and learning consequences that follow. The proof (Appendix A) expands Λ to fourth order, takes the γ-second moment of the leading Hermite term by Wick's theorem, and bounds the remainder by the moment condition; the scalar leading term of Equation (4) is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$. The leading orders of a degree-2 and a degree-4 coordinate do not couple, because He_2 and He_4 are γ-orthogonal, so the efficient information of a shape coordinate after the nuisance coordinates are estimated has the same rate as its raw information.

Remark 4 (General order). The same grading gives, for a coordinate whose lowest non-vanishing Hermite footprint has degree m, $I_t(\theta) \propto e^{-2mt}$: the leading term of $\partial_\theta \log P_t$ is e^{-mt} times a degree-m Hermite polynomial, whose γ-second moment is a constant. The constants are explicit for $m = 2, 4$ above.

Corollary 5 (Sample complexity by order). Estimating a covariance coordinate to fixed precision from sequences observed at level t needs $M \propto e^{4t}$ of them; a shape coordinate needs $M \propto e^{8t}$. The ratio of the two informations is $\propto e^{-4t}$.

Where the asymptote begins. The expansion parameter is $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0)$, not e^{-2t}, with $\lambda_{\max} \le (1+\rho)/(1-\rho)$ for the chain. Below $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ the long-wavelength covariance modes are still of order one and the rates are shallower; above it the exponents of Theorem 3 hold. At $\rho = 0.85$, $t_* = 1.17$: a strongly correlated chain keeps its correlation information longer, and any measurement stopped near t_* sees slopes short of -4 and -8.

Verification with the exact score. Figure 2 tests Theorem 3 where nothing has to be approximated. For the Gaussian chain the closed form gives slope -3.97 above t_* at $\rho = 0.85$ and the scaled information 0.24, 0.69, 0.95 at $t = 1, 2, 3$. For the non-Gaussian chain the shape information, estimated from per-sequence scores obtained by Fisher's identity through the tree-exact recursion, falls with slope -7.94 ($\beta = 2$) and -8.01 ($\beta = 3$) above t_*, against the predicted -8, while the correlation information falls with slope -3.98 against -4; the kinked Laplace density ($\beta = 1$) reaches the quadrature floor of the outer-product estimate near $t = 2.5$. The closure gap on the Laplace chain decays with slope -7.93 and reproduces the constant $\frac{1}{6}\|\kappa_4\|_F^2$ (113.0 for $L = 32$) to 0.99 at $t = 3$. The theorem also explains a measurement the previous literature on this model reported as a discrepancy: information slopes of about -3 and -6 obtained over $t \le 1.6$ are the pre-asymptotic values below $t_* = 1.17$, not a failure of the e^{-4t}, e^{-8t} law.

What this says about score learning. An estimator that carries only second-order structure cannot go below the LMMSE floor $\|S - S^{\mathrm{G}}\|$, which on the Laplace chain at $\rho = 0.85$ is 0.168 of the score energy pooled over the twelve-level schedule and 0.28 at the lowest level (Appendix D). A learner that recovers the fourth-order structure has, by Corollary 5, e^{4t} fewer effective samples per sequence at level t than one that recovers only covariance: the twelve-level schedule is, for shape, effectively a four-level one. Both facts organise the experiments of Section 5.

4. Learning the dynamics through noise

Now drop the assumption that K is known. We observe M noised sequences $\mathbf{x}^\mu$, each at one level t_μ, never the clean chain, and maximise the marginal log-likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_t[t_\mu, \theta](\mathbf{x}^\mu)$ of a parameterised kernel K_θ.

The exact gradient from one sweep. Because only the transition factors depend on θ, Fisher's identity (Cappé et al., 2005) gives, on the grid,

$$\nabla_\theta \mathcal{L}(\theta) = \big\langle \mathbf{\Xi}(\theta),\, \nabla_\theta \log K_\theta \big\rangle, \qquad \mathbf{\Xi}_{kl} = \sum_\mu \sum_{i \ge 2} P(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu), \tag{7}$$

where the expected transition mass $\mathbf{\Xi}$ is read off the pairwise marginals that the same forward–backward pass returns. No differentiation through the recursion is needed, and the identity is verified against finite differences to 10^{-9} (Appendix C). $\mathbf{\Xi}$ is also all the M-step of EM needs: for a Gaussian kernel the maximiser is the belief-weighted Yule–Walker pair $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$ with $S_{ab} = \sum_{kl} \mathbf{\Xi}_{kl} u_k^a u_l^b$; for the mixture-innovation kernel $K_\theta(a' \mid a) = \sum_{c=1}^{C} \pi_c \mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)$ used below, the complete-data problem carries a second latent label and we take conditional-maximisation sweeps, a generalised

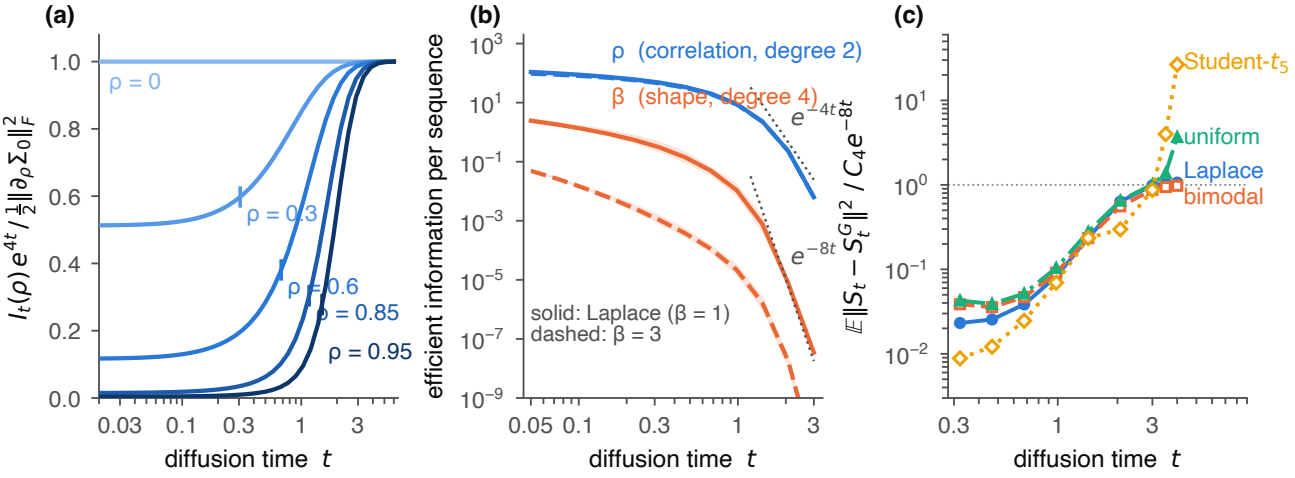


Figure 2. What survives diffusion noise. (a) Exact Fisher information of the Gaussian chain about ρ, scaled by its predicted asymptote $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$, for five ρ; every curve reaches 1, and the tick on each marks $t_*(\rho)$. (b) Efficient information per noised sequence about the correlation coordinate ρ (degree 2) and the shape coordinate β (degree 4) of the generalised-Gaussian chain, from tree-exact inference and Fisher's identity, 512 sequences $\times$ 8 replicates (bands: replicate range); guides e^{-4t} and e^{-8t}. Above t_* the fitted slopes are -4.00 (ρ) and -8.01 (β, at $\beta = 3$). (c) The price of second-order closure: $\mathbb{E}\|S - S^{\mathrm{G}}\|^2$ measured by tree-exact BP, divided by the predicted $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, for four innovation families. Laplace and the bimodal mixture reach the constant to within 0.99–0.98; Student-t_5, whose sixth moment is infinite, departs as the moment condition predicts; the uniform family reaches the floor of double precision near $t = 3.5$.

EM whose ascent is monotone (Dempster et al., 1977; Meng & Rubin, 1993). The estimator sees one frozen noised view per sequence and never a clean trajectory; each reported fit selects its iteration count on a validation bundle disjoint from the test bundle.

Shape is learned e^{4t} more slowly than correlation. Corollary 5 makes a prediction about learning that the exact score does not need: fitting from M sequences all observed at level t, the error of $\widehat{\rho}$ should grow like e^{2t} and the error of the fitted innovation excess kurtosis $\widehat{\kappa_4}$ like e^{4t} beyond t_*. Figure 3 measures both with $M = 512$ Laplace-chain sequences and 16 replicates per level. Figure 3 measures both with $M = 512$ Laplace-chain sequences at $\rho = 0.85$ and 16 replicates per level, against the information reference read from the exact-score computation of Section 3: the Cramér–Rao bound $1/\sqrt{M\, I_t}$ for the correlation and, through the chain rule at the Laplace member of the one-parameter shape family, for the excess kurtosis. The estimator sits at the bound: its error is 1.02× the bound for $\widehat{\rho}$ and 1.02× for $\widehat{\kappa_4}$, averaged over the levels at which either is informative. What the channel does is therefore read off the bound itself. The shape bound grows as e^{4t} and crosses the full range of the coordinate (3) at $t = 0.98$; at $t = 1$ the fitted excess kurtosis is 2.794 from the truth, the fit having collapsed to a Gaussian innovation, while $\widehat{\rho}$ is still within 0.0092. Correlation is lost only past the crossover $t_* = 1.17$: 0.0325 at $t = 1.4$ and 0.3430 at $t = 2$, where the estimator also leaves the bound. Below the crossover both errors grow more slowly than the asymptotic rates, as Section 3 predicts (fitted log-slopes 1.29 and 3.03 on $[0.2, 0.7]$ against 2 and 4); the asymptotic window itself lies beyond the reach of $M = 512$ sequences at this correlation, because the shape estimate saturates before t_*. At $t = 0.1$ the transition law is recovered to Hellinger distance 0.048, and the clean-data fit gives the optimiser's own floor, 0.0029 and 0.310. This is the mechanism behind the convergence asymmetry reported for this model—correlation settles in tens of EM updates, shape in hundreds—and behind the one-view protocol's arithmetic: of twelve levels spread over $t \in [0.05, 3]$, only the lowest four carry appreciable information about shape.

5. What explicit structure buys, and where it stops

All comparisons use the frozen configuration of Appendix C: Laplace chain, $\rho = 0.85$, $L = 32$, twelve log-spaced levels on $[0.05, 3]$, and the schedule-pooled relative score error against the tree-exact reference (squared error and reference energy summed over levels before the square root). Sixteen paired seeds are the inferential unit; architectures and hyperparameters are selected on two development seeds that never enter a reported number, and every arm's budget is selected on a per-seed validation bundle disjoint from the test bundle. Three reference points bracket ev-

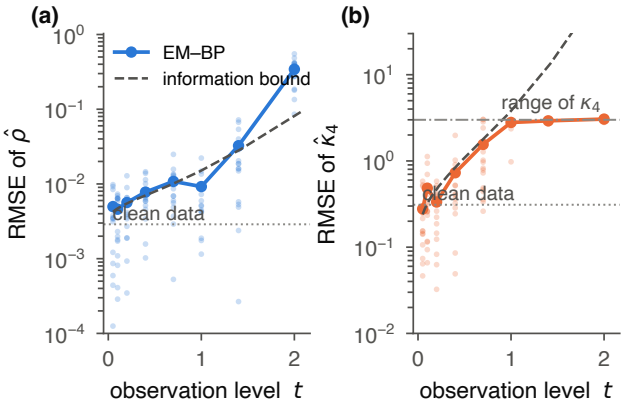


Figure 3. Learning through noise. Recovery of the correlation coordinate and of the innovation shape from $M = 512$ sequences observed at a single level t, sixteen replicates; the predicted growth e^{2t} and e^{4t} of the two errors beyond t_* (dotted).

Table 1. Schedule-pooled relative score error on all sites, mean over 16 seeds (s.e. in the text macros; the isotropic reference is 0.669 at every size). Every arm is scored on identical test bundles. The last column is the same capacity fitted to the innovation law itself—the C-component mixture closest in Kullback–Leibler divergence, an infinite-data reference for the family rather than a bound; † marks fits whose narrowest component is under two grid cells.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022†
$C = 16$	0.100	0.056	0.036	0.022	0.0007†

ery estimator (Appendix D): the isotropic score $-\mathbf{x}$ at 0.670, the LMMSE oracle with the true covariance at 0.168—the floor of every second-order estimator, by Proposition 1—and the exact score at zero.

Capacity: the margin is fourth-order. Table 1 places every arm on identical test bundles. A Gaussian-innovation kernel ($C = 1$) lands on the LMMSE floor at every size (0.174, 0.168, 0.168, 0.167 against the floor 0.167–0.167), as Proposition 1 requires; the mixture kernel goes far below it, reaching 0.023 at $M = 2048$, 7.9× under the floor. The paired $C = 8$ versus $C = 1$ difference is 0.144 (bootstrap $[0.141, 0.147]$) at $M = 2048$ and 0.075 ($[0.059, 0.090]$) already at $M = 32$, in favour of the mixture in 16 and 16 of 16 seeds. At $M = 2048$ the $C = 8$ arm still selects the top of its 1200-iteration ladder in 10 of 16 seeds, so its error there is an upper bound on what a longer ladder would give. The fitted innovation excess kurtosis is 2.89 at $M = 2048$ against the true 3; 1 of 320 fitted cells fail the two-grid-cell resolution gate and are reported, not dropped. The last row of Table 1 is the family's infinite-data reference: the C-component mixture closest in Kullback–Leibler divergence to the Laplace innovation, scored on the same bundles. Its $C = 1$ member is the covariance twin and scores 0.167, the LMMSE floor to three decimals, as Proposition 1 requires; at $C = 8$ it scores 0.0022 (a fit whose narrowest component is under two grid cells, as it is for every $C \geq 8$), so the fitted kernel at $M = 2048$ sits 10.3× above what its own family can express: a data limit, not a capacity limit. The reference is not a bound—a Kullback–Leibler-optimal innovation need not minimise the score error—and the best capacity at each size, $C = 2$ at $M = 32$ rising to $C = 16$ at $M = 2048$, is the usual variance-to-bias trade. A Gaussian-innovation kernel ($C = 1$) lands on the LMMSE floor, as Proposition 1 requires; the mixture kernel goes far below it. An earlier study of this model concluded the opposite from a sweep whose single-Gaussian arm alone stopped at its validation optimum while richer arms ran to a fixed cap; with the checkpoint selected identically for every capacity the effect is unambiguous (Appendix F). Moving the innovation from Laplace (excess kurtosis +3) through the Gaussian to the platykurtic end at fixed ρ and variance (Figure 4), the LMMSE floor traces a V: 0.166 at the Laplace end, exactly zero at $\beta = 2$ where the covariance twin is the truth, and 0.119 at the platykurtic

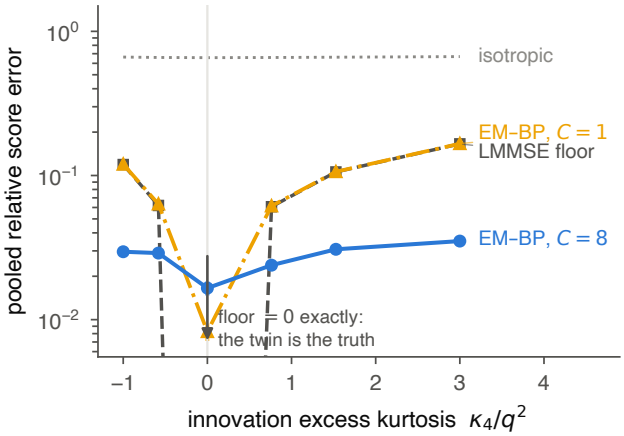


Figure 4. Capacity buys exactly what the closure discards. Pooled relative score error against the innovation's excess kurtosis for the generalised-Gaussian family at fixed ρ and variance ($M = 512$, 16 seeds, ± s.e.): the Gaussian-innovation kernel tracks the LMMSE floor, which is the exact price of second-order closure for each member; the mixture kernel tracks the exact score, and the two meet at the Gaussian member, where there is nothing beyond second order to learn.

end. The Gaussian-innovation kernel follows that V within its own standard error at every member, which is Proposition 1 read across a family rather than at a point; the mixture kernel stays between 0.017 and 0.035 throughout. The paired gain of $C = 8$ over $C = 1$ is therefore ordered by κ_4: 0.131 ($[0.127, 0.135]$) at Laplace, 0.075 at $\kappa_4 = +1.53$, and 0.090 at the platykurtic end. At the Gaussian member the gain reverses: -0.008 ($[-0.011, -0.005]$), a small but unambiguous loss, because there is no fourth-order structure to buy and the extra components only add estimation variance. Non-Gaussian capacity is worth what Theorem 3 says it is worth—the fourth-order structure the covariance twin leaves behind—and nothing where the twin leaves nothing.

Baselines under one recipe. Table 2 compares six architectures trained under one recipe (per-example noise level, batch 64, Adam with clipping, both output parameterisations, checkpoint selected on validation), each with the configuration that won its family's screen on the development seeds (Table 4, Appendix C), on the test bundles of the capacity experiment. Three things separate them. Weight sharing separates the fully connected network from the rest at every size (0.517 against 0.186 at $M = 32$). Locality is the better prior when sequences are scarce—the window head is the best network at $M = 32$—but it is a ceiling rather than a prior once they are not: the window head moves only from 0.121 to 0.118 between $M = 512$ and $M = 2048$, while the global architectures keep improving, the best reaching 0.060. A radius-4 window cannot represent a message that has travelled the length of the chain, and the chain's exact score is exactly such a message.

Given enough clean sequences the networks pass the second-order floor: 5 of the six sit below 0.167 at $M = 2048$ (Proposition 1). Nothing intrinsic holds them at second order; they need data. The structured estimator is ahead at every size nonetheless—0.023 at $M = 2048$, a paired factor 2.83 ($[2.37, 3.42]$, favouring EM–BP in 16 of 16 seeds) below the best network—but that comparison is not information-matched: the networks see clean sequences and a fresh corruption at every gradient step, EM–BP one noised view per sequence and no clean chain at all.

The comparison at matched information. Trained at EM–BP's own interface, the same architectures lose most of their advantage from scale. The best matched arm at $M = 2048$ is the window head trained on one fixed corruption of a clean sequence, at 0.155 against EM–BP's 0.023 on the same bundles, a paired factor 7.3 ($[6.2, 8.7]$). Implicit score matching on the one-view bundle, which uses no clean target at all, is worse again (0.165 for the window head, 0.421 for the transformer). The gap between the standard-recipe rows and these is the value of clean data with fresh corruptions; the factor of 7.3 that remains is what the factorisation and the exact E-step are worth at equal information.

Length extrapolation. Fitted at $L = 32$ and evaluated against the exact score of longer chains (Table 3), the EM–BP kernel holds 0.023 at $L = 32$ and 0.023 at $L = 256$: the law is local, so inference at any length is the same exact recursion. The convolutional and recurrent networks improve slightly, as they should—longer chains have proportionally fewer boundary sites. The transformer with learned relative positions does not: it degrades from 0.063 to 0.703, a failure of the positional encoding beyond its training length rather than of the score it learned (Press et al., 2022; Kazemnejad et al., 2023).

Generation. [PENDING: E05 pending: reverse-SDE samples under the exact, EM–BP, LMMSE and network scores]

Misspecification. [PENDING: E08 pending: advantage against a misspecification parameter; break-even]

6. Related work

Solvable diffusion and the Gaussian part of the score. Diffusion and score matching are due to Sohl-Dickstein et al. (2015); Ho et al. (2020); Song et al. (2021) and Hyvärinen (2005); Vincent (2011); the score–posterior-mean identity is Tweedie's (Robbins, 1956; Efron, 2011). Exactly solvable targets have been Gaussian (Pierret & Galerne, 2024), mean-field or mixture-like (Biroli & Mézard, 2023; Biroli et al., 2024; Cui et al., 2025), or hierarchical with discrete leaves (Sclocchi et al., 2025), on which U-Nets and transformers implement belief propagation (Mei, 2024; Mei & Wu, 2023; Garnier-Brun et al., 2024). A recurrent empirical observation is that learned scores are dominated by their Gaussian part over a wide noise range (Wang & Vastola, 2024; Li et al., 2024; Kadkhodaie et al., 2024; Wang et al., 2026), and that downstream statistics depend on the first two moments alone (Ghane et al., 2025). Our chain is the degenerate tree of depth L with continuous non-Gaussian states; the Gaussian part of its score is the Kalman smoother of the covariance twin (Proposition 1), and Equation (4) is the sharp rate, with its constant and crossover, behind the dominance those papers report.

Table 2. All-site schedule-pooled relative score error, mean over the confirmatory seeds (± s.e. in the macros), every arm on identical test bundles. Oracles and EM–BP ($C = 8$) from the capacity experiment; networks under the common recipe with the screened configuration.

M	isotropic	LMMSE floor	MLP (global)	window head	dilated TCN	bidirectional GRU	transformer	1-D U-Net	EM
32	0.669	0.167	0.517	0.186	0.223	0.201	0.231	0.227	0.
128	0.669	0.167	0.391	0.141	0.139	0.121	0.151	0.150	0.
512	0.669	0.167	0.276	0.121	0.096	0.078	0.097	0.095	0.
2048	0.669	0.167	0.208	0.118	0.068	0.060	0.063	0.066	0.

Table 3. Length extrapolation at $M = 2048$: pooled relative score error of estimators fitted at $L = 32$ and evaluated against the tree-exact score of longer chains, mean over 16 seeds. Longer chains are slightly easier (proportionally fewer boundary sites), so a length-agnostic arm should improve a little; the transformer instead breaks down beyond its training length.

arm	M	$L = 32$	$L = 64$	$L = 128$	$L = 256$
bidirectional GRU	2048	0.060	0.057	0.057	0.056
EM–BP ($C = 8$)	2048	0.023	0.023	0.023	0.023
dilated TCN	2048	0.068	0.066	0.065	0.064
transformer	2048	0.063	0.243	0.514	0.703
1-D U-Net	2048	0.066	0.064	0.062	0.062
window head	2048	0.118	0.099	0.088	0.082

What survives noise: the low-SNR literature. Theorem 3 belongs to the low-SNR expansions of estimation theory. In Gaussian channels the mutual information and the MMSE expand in the SNR with fourth-order statistics entering at second order (Prelov & Verdú, 2004; Guo et al., 2005; Dytso et al., 2023; Kong et al., 2023); for latent-variable models the log-likelihood expands with the order-k term matching k-th moments (Katsevich & Bandeira, 2023), which yields the σ^{2k} sample complexity of multireference alignment and orbit recovery (Perry et al., 2019; Bandeira et al., 2023) and, most closely, the layered Fisher-information geometry of Balanov et al. (2026), where information about directions first identifiable at moment order k scales as SNR^k. Under the OU channel $\mathrm{SNR}_t \simeq e^{-2t}$, so that statement and our e^{-2mt} are the same law in different clocks. Aarts et al. (2025) write the forward-process cumulant attenuation e^{-nt} and show that score networks regenerate the cumulants. What is not in that literature, and is what this paper adds, is the score-level statement for a Markov (dependent) rather than i.i.d. latent structure, its constant $\frac{1}{6}\|\kappa_4\|_F^2$ and crossover t_*, the exact verification on a chain, and the learning consequences with an exact E-step. Speciation and collapse times (Biroli et al., 2024; Achilli et al., 2026) and the information-restricted Bayesian analysis of Hunt et al. (2026) are second-moment phenomena in this vocabulary; heavy-tailed forward processes (Pandey et al., 2024; Tang et al., 2026) change the channel and therefore the ladder.

Learning from noisy data only. Ambient diffusion learns a clean distribution from corrupted samples by predicting a further corruption (Daras et al., 2023; 2024); EM with diffusion posterior sampling in the E-step (Rozet et al., 2024; Bai et al., 2024) and SURE-type surrogates (Kawar et al., 2024) address the same problem nonparametrically. Our learner is parametric and its E-step is exact, so the question changes from "can the clean law be recovered" to "how fast is it lost", which Theorem 3 answers order by order. The estimator is classical: sum-product on a chain is forward–backward (Baum et al., 1970; Rabiner, 1989), grid quadrature is Kitagawa's smoother (Kitagawa, 1987), Fisher's identity and EM are Dempster et al. (1977); Cappé et al. (2005), and the Gaussian special case is the Shumway–Stoffer algorithm (Shumway & Stoffer, 1982). Consistency of the maximum-likelihood estimator for general-state hidden Markov models is Douc et al. (2011); the identifiability of a Markov signal observed through independent noise, which is what makes the one-view protocol of Section 4 well posed, is Gassiat et al. (2020). Spectral and moment methods for hidden Markov models (Hsu et al., 2012) are the natural high-noise competitor that Corollary 5 suggests.

Sequence denoisers and length. The baselines of Section 5 are the standard sequence architectures (Bai et al., 2018; Vaswani et al., 2017; Ronneberger et al., 2015); that attention-based length extrapolation turns on the positional encoding (Press et al., 2022; Kazemnejad et al., 2023) is visible in Table 3, here against an exact reference at every length.

7. Limitations and conclusion

The model is a chain in one dimension with a linear autoregressive form; the theorem holds for any law with the stated moments, but the exact verification and the learning results are for chains. The estimator's advantage is the correctness of its factorisation: under a non-Markov truth it is misspecified in a way

more data does not repair, and Section 5 locates where that begins. Inference costs $O(LN_g^2)$ per sequence, far above a network forward pass, so the advantage is in data and understanding, not in wall-clock. The comparison with denoisers equalises the number of clean sequences under each method's standard recipe and, separately, matches the information interface; it does not rank learning algorithms in general.

What the laboratory teaches is general: diffusion noise erases statistical structure order by order, at rates fixed by Hermite degree, so a score learner that reaches only second order is bounded away from the truth by an e^{-8t}-scaled term and a learner of shape pays e^{4t} in samples per level. Explicit structure is the way to recover what the data still carry; the price of the wrong structure is a bias that no sample size removes.

Impact statement

This paper presents theoretical and empirical work on how diffusion noise erases statistical structure and on learning structured dynamics from noised data, evaluated on synthetic models with known ground truth. Its societal consequences are those of a better understanding of generative diffusion models in general; we see no specific ethical concern that requires discussion here.

References

Aarts, G., Habibi, D. E., Wang, L., and Zhou, K. On learning higher-order cumulants in diffusion models. Machine Learning: Science and Technology, 6: 025004, 2025. arXiv:2410.21212.

Achilli, B., Benedetti, M., Biroli, G., and Mézard, M. Theory of speciation transitions in diffusion models with general class structure, 2026.

Bai, S., Kolter, J. Z., and Koltun, V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, 2018. arXiv:1803.01271.

Bai, W., Wang, Y., Chen, W., and Sun, H. An expectation-maximization algorithm for training clean diffusion models from corrupted observations. In Advances in Neural Information Processing Systems, 2024. arXiv:2407.01014.

Balanov, A., Bendory, T., and Edidin, D. The generalized method of moments is (almost) statistically efficient in low-SNR Gaussian latent-variable models, 2026. arXiv:2605.30095.

Bandeira, A. S., Blum-Smith, B., Kileel, J., Perry, A., Niles-Weed, J., and Wein, A. S. Estimation under group actions: recovering orbits from invariants. Applied and Computational Harmonic Analysis, 66: 236–319, 2023.

Baum, L. E., Petrie, T., Soules, G., and Weiss, N. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. The Annals of Mathematical Statistics, 41(1):164–171, 1970.

Biroli, G. and Mézard, M. Generative diffusion in very large dimensions. Journal of Statistical Mechanics: Theory and Experiment, 2023(9):093402, 2023.

Biroli, G., Bonnaire, T., de Bortoli, V., and Mézard, M. Dynamical regimes of diffusion models. Nature Communications, 15:9957, 2024. doi: 10.1038/s41467-024-54281-3.

Cappé, O., Moulines, E., and Rydén, T. Inference in Hidden Markov Models. Springer, 2005.

Cui, H., Pehlevan, C., and Lu, Y. M. A solvable model of learning generative diffusion: theory and insights, 2025.

Daras, G., Shah, K., Dagan, Y., Gollakota, A., Dimakis, A. G., and Klivans, A. Ambient diffusion: learning clean distributions from corrupted data. In Advances in Neural Information Processing Systems, 2023. arXiv:2305.19256.

Daras, G., Dimakis, A. G., and Daskalakis, C. Consistent diffusion meets Tweedie: training exact ambient diffusion models with noisy data. In International Conference on Machine Learning, 2024. arXiv:2404.10177.

Dempster, A. P., Laird, N. M., and Rubin, D. B. Maximum likelihood from incomplete data via the EM algorithm. Journal of the Royal Statistical Society, Series B, 39(1):1–38, 1977.

Douc, R., Moulines, E., Olsson, J., and van Handel, R. Consistency of the maximum likelihood estimator for general hidden Markov models. Annals of Statistics, 39(1):474–513, 2011.

Dytso, A., Poor, H. V., and Shamai, S. Conditional mean estimation in Gaussian noise: a meta derivative identity with applications. IEEE Transactions on Information Theory, 69:1883–1910, 2023.

Efron, B. Tweedie's formula and selection bias. Journal of the American Statistical Association, 106 (496):1602–1614, 2011.

Garnier-Brun, J., Mézard, M., Moscato, E., and Saglietti, L. How transformers learn structured data: insights from hierarchical filtering, 2024.

Gassiat, E., Le Corff, S., and Lehéricy, L. Identifiability and consistent estimation of nonparametric translation hidden Markov models with general state space. Journal of Machine Learning Research, 21(115):1–40, 2020.

Ghane, R., Bao, A., Akhtiamov, D., and Hassibi, B. Gaussian universality for diffusion models, 2025. arXiv:2501.07741.

Guo, D., Shamai, S., and Verdú, S. Mutual information and minimum mean-square error in Gaussian channels. IEEE Transactions on Information Theory, 51(4):1261–1282, 2005. doi: 10.1109/TIT.2005.844072.

Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In Advances in Neural Information Processing Systems (NeurIPS), volume 33, pp. 6840–6851, 2020.

Hsu, D., Kakade, S. M., and Zhang, T. A spectral algorithm for learning hidden Markov models. Journal of Computer and System Sciences, 78(5):1460–1480, 2012.

Hunt, H., Kamb, M., and Ganguli, S. An exact information theory of generalization phase transitions in Bayesian diffusion models, 2026. arXiv:2607.08041.

Hyvärinen, A. Estimation of non-normalized statistical models by score matching. Journal of Machine Learning Research, 6:695–709, 2005.

Kadkhodaie, Z., Guth, F., Simoncelli, E. P., and Mallat, S. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In International Conference on Learning Representations, 2024. Outstanding Paper award.

Katsevich, A. and Bandeira, A. S. Likelihood maximization and moment matching in low SNR Gaussian mixture models. Communications on Pure and Applied Mathematics, 76(4):788–842, 2023. arXiv:2006.15202.

Kawar, B., Elata, N., Michaeli, T., and Elad, M. GSURE-based diffusion model training with corrupted data. Transactions on Machine Learning Research, 2024. arXiv:2305.13128.

Kazemnejad, A., Padhi, I., Natesan Ramamurthy, K., Das, P., and Reddy, S. The impact of positional encoding on length generalization in transformers. In Advances in Neural Information Processing Systems, 2023. arXiv:2305.19466.

Kitagawa, G. Non-Gaussian state-space modeling of nonstationary time series. Journal of the American Statistical Association, 82(400):1032–1041, 1987.

Kong, X., Brekelmans, R., and Ver Steeg, G. Information-theoretic diffusion. In International Conference on Learning Representations, 2023. arXiv:2302.03792.

Kschischang, F. R., Frey, B. J., and Loeliger, H.-A. Factor graphs and the sum–product algorithm. IEEE Transactions on Information Theory, 47(2): 498–519, 2001.

Li, X., Dai, Y., and Qu, Q. Understanding generalizability of diffusion models requires rethinking the hidden Gaussian structure. In Advances in Neural Information Processing Systems, 2024. arXiv:2410.24060.

Mei, S. U-nets as belief propagation: Efficient classification, denoising, and diffusion in generative hierarchical models, 2024. arXiv:2404.18444.

Mei, S. and Wu, Y. Deep networks as denoising algorithms: sample-efficient learning of diffusion models in high-dimensional graphical models, 2023. arXiv:2309.11420; IEEE Transactions on Information Theory, Xplore document 10857317.

Meng, X.-L. and Rubin, D. B. Maximum likelihood estimation via the ECM algorithm: A general framework. Biometrika, 80(2):267–278, 1993.

Pandey, K., Pathak, J., Xu, Y., Mandt, S., Pritchard, M., Vahdat, A., and Mardani, M. Heavy-tailed diffusion models, 2024. arXiv:2410.14171.

Pearl, J. Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference. Morgan Kaufmann, San Mateo, CA, 1988.

Perry, A., Weed, J., Bandeira, A. S., Rigollet, P., and Singer, A. The sample complexity of multireference alignment. SIAM Journal on Mathematics of Data Science, 1(3):497–517, 2019.

Pierret, E. and Galerne, B. Diffusion models for Gaussian distributions: exact solutions and Wasserstein errors, 2024. arXiv:2405.14250.

Prelov, V. V. and Verdú, S. Second-order asymptotics of mutual information. IEEE Transactions on Information Theory, 50(8):1567–1580, 2004.

Press, O., Smith, N. A., and Lewis, M. Train short, test long: attention with linear biases enables input length extrapolation. In International Conference on Learning Representations, 2022. arXiv:2108.12409.

Rabiner, L. R. A tutorial on hidden Markov models and selected applications in speech recognition. Proceedings of the IEEE, 77(2):257–286, 1989.

Robbins, H. An empirical Bayes approach to statistics. In Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, volume 1, pp. 157–163, 1956.

Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 234–241, 2015.

Rozet, F., Andry, G., Lanusse, F., and Louppe, G. Learning diffusion priors from observations by expectation maximization. In Advances in Neural Information Processing Systems, 2024. arXiv:2405.13712.

Sclocchi, A., Favero, A., and Wyart, M. A phase transition in diffusion models reveals the hierarchical nature of data. Proceedings of the National Academy of Sciences, 122(1):e2408799121, 2025.

Shumway, R. H. and Stoffer, D. S. An approach to time series smoothing and forecasting using the EM algorithm. Journal of Time Series Analysis, 3(4): 253–264, 1982.

Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., and Ganguli, S. Deep unsupervised learning using nonequilibrium thermodynamics. In Proceedings of the 32nd International Conference on Machine Learning (ICML), pp. 2256–2265, 2015.

Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In International Conference on Learning Representations (ICLR), 2021.

Tang, W., Touzi, N., Zhang, Z., and Zhou, X. Y. Tweedie's formulae and diffusion generative models beyond Gaussian, 2026. arXiv:2605.19391.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS), volume 30, 2017.

Vincent, P. A connection between score matching and denoising autoencoders. Neural Computation, 23(7): 1661–1674, 2011.

Wang, B. and Vastola, J. J. The unreasonable effectiveness of Gaussian score approximation for diffusion models and its applications. Transactions on Machine Learning Research, 2024. arXiv:2412.09726.

Wang, B., Zavatone-Veth, J., and Pehlevan, C. A random matrix theory perspective on the consistency of diffusion models. In International Conference on Machine Learning, 2026. arXiv:2602.02908.

A. Proofs

A.1. Proposition 1

Write $\overrightarrow{m}_i^{\mathrm{P}}$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood. We show by induction that every projected message equals the corresponding exact message of the covariance-matched Gaussian chain. The initial message is Gaussian or projected to the Gaussian with the same first two moments, which is the Gaussian chain's initial message. If $\overrightarrow{m}_{i-1}^{\mathrm{P}}$ is that chain's message, with mean m and variance v, multiplying by ℓ_{i-1} gives a Gaussian with the precision-weighted moments, identical in both chains; the transition integral against $a_i = \rho a_{i-1} + \varepsilon_i$ is a convolution whose mean and variance are $(\rho m, \rho^2 v + q)$ for every innovation with moments $(0, q)$, so Π_{G} returns the Gaussian chain's next message. The backward sweep runs the same argument with the inverted map $(m, v) \mapsto (m/\rho, (v+q)/\rho^2)$, which is where $\rho \neq 0$ is used, starting from the terminal likelihood ℓ_L (the flat terminal message is not normalisable and contributes no factor). Every projected belief is therefore the Gaussian chain's belief, whose posterior mean is linear in $\mathbf{x}$ and coincides with the LMMSE estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Since the recursion never read φ beyond $(0, q)$, the estimator is the same for every such law. The algebraic steps (the two moment maps as integrals, the covariance identities, and the optimality of the normal-equation gain in arbitrary dimension) are machine-checked in Lean 4 (lean/README.md); the projection step itself is not.

A.2. Theorem 2

For $z \sim \mathcal{N}(0,1)$ and real a, α with $\beta = \sqrt{1-\alpha^2}$, the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$ evaluated at $x = \alpha a + \beta z$ and averaged over z gives $e^{s\alpha a - s^2\alpha^2/2} = \sum_n \mathrm{He}_n(a)(\alpha s)^n/n!$, hence $\mathbb{E}_z \mathrm{He}_n(\alpha a + \beta z) = \alpha^n \mathrm{He}_n(a)$. The coordinates of $\mathbf{z}$ are independent given $\mathbf{a}$, so the product form holds for H_ν; integrating over $\mathbf{a} \sim P_0$ requires only $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$. The cases $n \leq 4$ are verified in Lean as polynomial identities under the first four Gaussian moments.

A.3. Theorem 3

Exact identities. With $\alpha = e^{-t}$ and $\Delta_t = 1 - \alpha^2$, Bayes' rule gives $P_t(\mathbf{x}) = \gamma_{\Delta_t}(\mathbf{x})\, \mathbb{E}_{P_0} \exp\!\big(\alpha \mathbf{x}\cdot\mathbf{a}/\Delta_t - \alpha^2|\mathbf{a}|^2/2\Delta_t\big)$, where γ_{Δ_t} is the $\mathcal{N}(0, \Delta_t \mathbf{I})$ density. Hence $\log P_t(\mathbf{x}) = \log \gamma_{\Delta_t}(\mathbf{x}) + \Lambda(s, b)$ with $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and $\Lambda(s,b) = \log \mathbb{E}_{P_0} \exp(s\cdot\mathbf{a} - b|\mathbf{a}|^2/2)$. For $b > 0$ the tilted measure is integrable for every s, so Λ is real-analytic in (s, b) on $\mathbb{R}^L \times (0, \infty)$, and its Taylor coefficients at the origin are the joint cumulants of the linear form $s\cdot\mathbf{a}$ and the quadratic form $|\mathbf{a}|^2/2$; $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = \nabla_s \Lambda$ and $S = -\mathbf{x}/\Delta_t + (\alpha/\Delta_t)\nabla_s \Lambda$.

Grading. Expand Λ in (s, b) at the origin and grade a monomial $s^j b^k$ by $j + 2k$; since $s = O(\alpha)$ and $b = O(\alpha^2)$, the grade is the order in α. By the symmetry of P_0 under $\mathbf{a} \mapsto -\mathbf{a}$ only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$, with $\Lambda_2 = \frac{1}{2} s^\top \mathbf{\Sigma}_0 s - \frac{b}{2} \operatorname{tr} \mathbf{\Sigma}_0$ and

$$\Lambda_4 = \tfrac{1}{24}\kappa_4[s,s,s,s] - \tfrac{b}{4}\sum_i \operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i, a_i) + \tfrac{b^2}{8}\sum_{ij} \operatorname{cum}(a_i, a_i, a_j, a_j),$$

where cum is the joint cumulant under P_0. Every fourth-order joint cumulant of linear forms in $\mathbf{a}$ equals its value under the covariance twin plus a linear contraction of κ_4, so $\Lambda_4 - \Lambda_4^{\mathrm{G}}$ is linear in κ_4 and $\Lambda_2 = \Lambda_2^{\mathrm{G}}$. The remainder R_6 is the Taylor remainder of grade six; on $\{|\alpha\mathbf{x}| \leq 1\}$ it is bounded by $C\alpha^6(1 + |\mathbf{x}|)^6$ with C depending on the moments of P_0 up to order eight.

Leading term of the score gap. $S - S^{\mathrm{G}} = (\alpha/\Delta_t)\nabla_s(\Lambda - \Lambda^{\mathrm{G}}) = (\alpha/\Delta_t)\big[\nabla_s(\Lambda_4 - \Lambda_4^{\mathrm{G}}) + \nabla_s R_6 - \nabla_s R_6^{\mathrm{G}}\big]$. Substituting $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and expanding in α, the grade-four term is $\alpha^4 \nabla_{\mathbf{x}}\big[\frac{1}{24}\sum_{ijkl}\kappa_4[ijkl] :x_i x_j x_k x_l:\big] + O(\alpha^6(1+|\mathbf{x}|)^5)$, where $:\cdot:$ is the Wick (Hermite) product: the three contributions of Λ_4 combine into the Hermite form because $b = \alpha^2(1 + O(\alpha^2))$ and the cumulant contractions with δ_{ii} are exactly the terms that convert monomials into Hermite polynomials. The m-th component is $\alpha^4 \frac{1}{6}\sum_{jkl}\kappa_4[mjkl] :x_j x_k x_l:$.

Second moment. Under γ, Wick's theorem gives $\mathbb{E}_\gamma[:x_j x_k x_l::x_{j'} x_{k'} x_{l'}:]$ equal to the number of perfect matchings of $\{j,k,l\}$ with $\{j',k',l'\}$ respecting indices; contracted with the symmetric tensor this yields

$\mathbb{E}_\gamma\big[(\frac{1}{6}\sum_{jkl}\kappa_4[mjkl]{:}x_jx_kx_l{:})^2\big] = \frac{3!}{36}\sum_{jkl}\kappa_4[mjkl]^2$, and summing over m, $\mathbb{E}_\gamma\|\text{leading term}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2\alpha^8$. Replacing $\mathbb{E}_\gamma$ by $\mathbb{E}_{P_t}$ costs a factor $1+O(\alpha^2\lambda_{\max})$: on the bulk, $P_t/\gamma = 1+\alpha^2\frac{1}{2}\langle\mathbf{\Sigma}_0-\mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + O(\alpha^4\lambda_{\max}^2(1+|\mathbf{x}|)^4)$, and the degree-2 correction has zero γ-mean against the square of a degree-3 polynomial up to terms of relative size $\alpha^2\lambda_{\max}$. The cross term between the leading term and the grade-six remainder is $O(\alpha^{10}\lambda_{\max})$ by Cauchy–Schwarz and the moment bound.

Tail. On $\{|\alpha\mathbf{x}| > 1\}$ the expansion is not used. There $|S - S^{\mathrm{G}}| \le 2|\mathbf{x}|/\Delta_t + \alpha(\mathbb{E}[|\mathbf{a}|\,|\mathbf{x}] + |\mathbb{E}^{\mathrm{G}}[\mathbf{a}|\mathbf{x}]|)/\Delta_t \le C(1+|\mathbf{x}|)/\Delta_t$ because the tilted mean is bounded by a polynomial in $|\mathbf{x}|$ under the moment condition, while $P_t(|\alpha\mathbf{x}| > 1) \le \exp(-c/\alpha^2)$ since $\mathbf{x}_t$ is a Gaussian convolution with variance at most one per coordinate. The tail contribution is therefore $O(\exp(-c'e^{2t}))$, below every power of α. Together, $\mathbb{E}_{P_t}\|S-S^{\mathrm{G}}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}(1+O(e^{-2t}\lambda_{\max}))$, which is Equation (4).

Information. $\partial_\theta \log P_t = \partial_\theta\Lambda$. If $\partial_\theta\mathbf{\Sigma}_0 \neq 0$ the grade-two term $\frac{\alpha^2}{2}\langle\partial_\theta\mathbf{\Sigma}_0, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle$ leads, and $\mathbb{E}_\gamma$ of its square is $\frac{\alpha^4}{4}\cdot 2\|\partial_\theta\mathbf{\Sigma}_0\|_F^2$ by Wick, giving Equation (5). If $\partial_\theta\mathbf{\Sigma}_0 = 0$ the grade-four term $\frac{\alpha^4}{24}\sum\partial_\theta\kappa_4[ijkl]{:}x_ix_jx_kx_l{:}$ leads, with γ-second moment $\frac{4!}{24^2}\|\partial_\theta\kappa_4\|_F^2\alpha^8$, giving Equation (6); the corrections and the tail are handled as above. Since ${:}\mathbf{x}\mathbf{x}^\top{:}$ and the degree-four Hermite polynomials are γ-orthogonal, the leading cross-information between a degree-2 and a degree-4 coordinate vanishes, so the leading term of the efficient information of the degree-4 coordinate equals that of its raw information. For the Gaussian chain $\mathbf{x}_t \sim \mathcal{N}(0, \mathbf{\Sigma}_t(\theta))$ exactly and the stated trace formula is the Gaussian Fisher information; its limit $e^{4t}I_t \to \frac{1}{2}\|\partial_\theta\mathbf{\Sigma}_0\|_F^2$ follows from $\mathbf{\Sigma}_t \to \mathbf{I}$ and $\partial_\theta\mathbf{\Sigma}_t = \alpha^2\partial_\theta\mathbf{\Sigma}_0$, and is machine-checked in Lean (lean/README.md, target 3).

The crossover. The expansion of P_t/γ used above is an expansion in $\alpha^2\mathbf{\Sigma}_0$, i.e. in $\alpha^2\lambda_{\max}(\mathbf{\Sigma}_0)$ in operator norm; for the chain $\lambda_{\max} \le (1+\rho)/(1-\rho)$ by the Toeplitz bound $\lambda_{\max} \le \sum_k \rho^{|k|}$. This is the source of the factor $\lambda_{\max}$ in every remainder and of $t_* = \frac{1}{2}\log\lambda_{\max}$; for the Gaussian chain the exact formula shows the same scale directly, since $\mathbf{\Sigma}_t^{-1}$ is within $\alpha^2\lambda_{\max}$ of $\mathbf{I}$ in operator norm.

B. Numerical representation and its control

Messages are represented on a uniform grid of N_{g} points on $[-A, A]$ with composite trapezoidal weights; the recursion is a scaled linear-domain forward–backward whose discarded constants are accumulated so the evidence $\log P_t(\mathbf{x})$ of the discretised model is exact to floating point. Three levels of exactness are kept apart: sum-product is exact at the level of functional messages; forward–backward is exact for the discretised factor model; and that model approximates the continuous chain with truncation (mass outside $[-A, A]$) and quadrature (second order in the spacing) errors that are measured, not assumed.

Validation where the answer is known. At $N_{\mathrm{g}} = 401$, $A = 8$ the recursion reproduces the Gaussian closed form and brute-force enumeration of the discretised posterior on short chains to double precision (tests/test_gaussian_bp_equivalence.py, tests/test_fisher_bruteforce.py). Against a finer reference ($N_{\mathrm{g}} = 1601$, $A = 12$) the worst relative score error over the twelve levels and four innovation families on ordinary draws is $9.9e-04$; under a three-standard-deviation displacement the heavy-tailed families lose two orders (1.7e-02). The worst forward-message boundary occupancy is $6.6e-05$ and the worst belief edge occupancy $2.9e-03$ over 60 production cells.

Refinement on the reported estimands. [PENDING: E04 pending: movement of the headline cells under refinement]

Floors in the theory verification. The outer-product information estimate of a shape coordinate is a small residual of order-one quantities; for the kinked Laplace density its quadrature floor is reached near $t = 2.5$ at $N_{\mathrm{g}} = 801$ (the replicate spread stops shrinking), which is why Figure 2b reports the asymptotic slope on the smooth members of the family. The closure gap of the uniform family reaches the double-precision floor of the BP means near $t = 3.5$, where its squared gap is 10^{-13} per chain.

Table 4. Architecture screen under one recipe on the development seeds (never used for a reported number): the best configuration per family, its parameter count, its pooled bulk risk over 4 development seeds, two sizes and 3 restarts of 32,000 steps per configuration, and the risk of the family's runner-up configuration.

architecture	selected configuration	parameters	screen risk	runner-up
MLP (global)	hidden: (128, 128), lr 0.001, eps	26,016	0.390	0.411
window head	radius: 4, width: 64, lr 0.001, eps	5,505	0.116	0.122
dilated TCN	channels: 32, dilations: (1, 2, 4, 8), lr 0.001, eps	17,057	0.143	0.160
bidirectional GRU	hidden: 32, lr 0.001, eps	15,169	0.127	0.133
transformer	d: 32, heads: 4, layers: 2, lr 0.003, eps	18,057	0.147	0.154
1-D U-Net	channels: 16, lr 0.001, eps	13,713	0.148	0.212

C. Experimental protocol and provenance

Frozen configuration. Laplace chain unless stated, $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0, 1)$, $L = 32$; grid $N_{\mathrm{g}} = 401$, $A = 8$; twelve log-spaced levels $t \in [0.05, 3.0]$; other families Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$), generalised Gaussian ($\beta \in [1, 6]$), all variance-matched. Sixteen confirmatory seeds 0–15; development seeds 10000–10001 for every architecture and hyperparameter choice (one 16,000-step run per configuration at $M \in \{32, 2048\}$, ranked by pooled bulk risk). Validation and test bundles are drawn per seed (1024 and 2048 sequences) from streams disjoint from every training stream; their hashes are stored with every row so pairing across arms is checkable.

Estimand. Schedule-pooled relative score error $\left(\sum_t \sum_{j,i} (\hat{s}_t(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i)^2 / \sum_t \sum_{j,i} s_t(\mathbf{x}_j)_i^2\right)^{1/2}$ on all sites (the headline), on the interior sites 5–28 (used for selection), and on the centre site (diagnostic). Ratios are formed within seed and averaged; intervals are percentile bootstraps over seeds.

Arms. EM–BP: mixture-innovation kernel, C as stated, ρ initialised at 0.3, one noised view per sequence with sequences split evenly across the twelve levels, checkpoint ladder $\{10, 20, 40, 60, 80, 120, 160, 220, 300, 400\}$ selected on the validation bundle. Networks: trained by denoising score matching on the clean sequences with the level and the corruption redrawn at every step, batch 64, Adam, gradient clipping at 1, both output parameterisations, checkpoint ladder to 32,000 steps selected on the validation bundle; the information-matched arm trains by implicit score matching on the same one-view noisy bundle as EM–BP and selects on the implicit objective of a held-out noisy bundle.

Provenance. Every run directory carries the commit, the dirty-tree list on the code path, the resolved configuration digest, the seeds, host and environment; a generator refuses to emit a table from a run whose code path was dirty, from an incomplete design, or from cells whose narrowest fitted component is under two grid cells. Every number in this paper is a macro emitted by such a generator; the manuscript check (tools/check_manuscript.py) fails the build on any typed decimal.

D. The oracle ladder

Table 5. Schedule-pooled relative score error of the two oracles against the tree-exact reference, 512 chains per family at the frozen configuration. The LMMSE column is the floor of every second-order estimator (Proposition 1).

family	excess kurtosis	isotropic $-\mathbf{x}$	LMMSE, true $\mathbf{\Sigma}_0$
Laplace	+3	0.670	0.168
Student-t_5	+6	0.668	0.158
uniform	−1.2	0.671	0.149
bimodal	−0.72	0.663	0.082
Gaussian	0	0.662	0.000

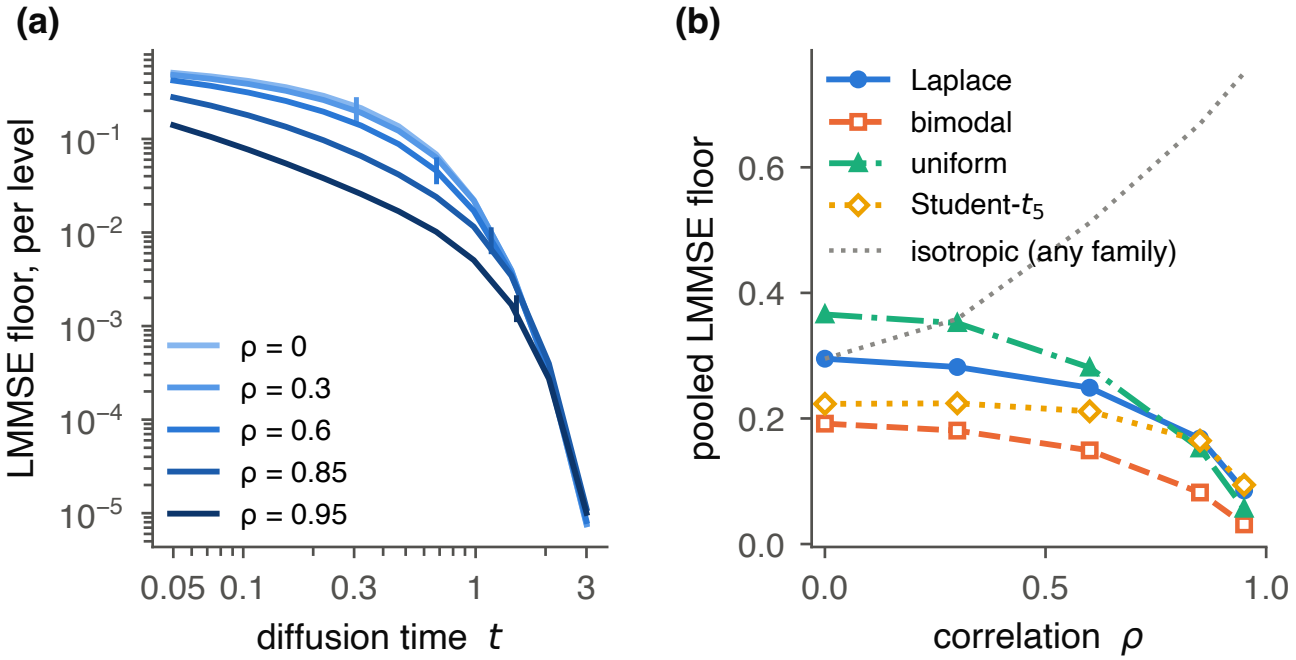


Figure 5. The second-order floor across correlation. (a) Per-level LMMSE floor on the Laplace chain for five ρ (ticks: $t_*(\rho)$); the floor collapses after the crossover. (b) Schedule-pooled floor against ρ for four innovation families, with the isotropic reference: strong correlation makes the score energy second-order-dominated, so the relative price of closure falls with ρ while the absolute fourth-order structure does not.

E. The second-order floor across correlation

The frozen configuration fixes $\rho = 0.85$. The price of second-order closure is a function of ρ as well as of the innovation law, and this appendix reports it over $\rho \in \{0, 0.3, 0.6, 0.85, 0.95\}$ for five families on the twelve-level schedule, computed from the exact score with no estimation involved.

Across correlation. The floor is a function of the innovation law and of ρ (Figure 5): on the Laplace chain it is 0.295 for independent sites, 0.168 at the frozen $\rho = 0.85$ and 0.085 at $\rho = 0.95$, where $t_* = 1.50$ and the covariance carries most of the score energy until late in the schedule. The families order by the size of their fourth cumulant at every ρ; the isotropic reference rises with ρ because the covariance itself is what an unstructured estimator must learn first.

F. Capacity: the previous reading and the stopping-rule artefact

An earlier study of this model compared innovation mixtures of $C \in \{1, 2, 4, 8, 16\}$ at $M \in \{128, 512\}$ over sixteen seeds and concluded that no capacity improved on a single Gaussian innovation, attributing the estimator's margin over the networks to the Markov factorisation and its covariance alone. Two facts overturn that reading. First, Proposition 1 bounds every Gaussian-innovation kernel by the LMMSE floor, which on the Laplace chain is 0.168 pooled over the schedule, whereas the same study's own headline estimator reported errors between two and nine times smaller. Second, its sweep selected only the initialisation on validation and ran every capacity to a fixed cap of 1200 iterations unless a 10^{-9} likelihood tolerance stopped it first; the tolerance fired in 31 of 32 single-Gaussian cells, at a median of 71 iterations, close to the validation optimum that the headline protocol selects for these sizes, and in none of the 96 cells with $C \geq 4$, which therefore reported checkpoints an order of magnitude past their optimum. "The single Gaussian is as good" was "the single Gaussian is the only arm that stopped in the right place". The capacity experiment of Section 5 selects the checkpoint identically for every capacity on a per-seed validation bundle and reports the untouched confirmatory seeds.

G. Aggregation robustness of the archival comparison

The archival comparison against the unstructured network (sixteen seeds, eight sizes) is retained here under its five aggregations, because the ratio of two errors is not invariant to the order of averaging; the schedule-pooled paired per-seed ratio is the one the text quotes.

H. Does the headline depend on how the ratio was averaged?

A ratio is a nonlinear function of two means, so its value depends on the order of averaging. the archival comparison (Appendix G) reports the schedule-pooled paired per-seed ratio: squared score error and reference score energy are summed over the twelve noise levels before the square root, the ratio is formed within each seed, and the seed is the inferential unit — the twelve levels share a training set and a fitted model. Being the definition stated in the text does not make it the only defensible summary, so here is the same data under four alternatives, with a paired interval on the one the table reports.

Table 6. The headline under five estimands. (0) is what the archival comparison (Appendix G) reports; (1) is the mean of per-(seed, level) ratios, the summary earlier drafts reported. The interval is a paired bootstrap over seeds, 20,000 resamples; seeds are resampled rather than cells, because cells within a seed share a fitted model.

M	(0) pooled	(1) per-cell	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (0)
32	6.9	7.0	6.8	6.1	6.8	[6.4, 7.5]
64	6.6	7.8	6.6	6.7	6.6	[5.6, 7.6]
128	7.9	8.8	8.0	7.8	8.6	[7.0, 8.8]
256	8.2	9.7	8.8	8.6	9.2	[7.3, 9.0]
512	8.2	9.0	8.6	7.9	8.2	[7.2, 9.2]
1024	9.0	12.5	9.7	10.8	11.1	[7.7, 10.4]
2048	8.6	15.5	10.0	13.2	12.9	[7.4, 9.9]
4096	10.9	17.6	12.4	15.6	16.0	[9.4, 12.5]

Two things follow.

The conclusion does not depend on the choice: no estimand at any size falls below 6.1, every bootstrap interval sits well clear of 1, and no aggregation produces a reversal anywhere.

But the estimands are far from interchangeable, and the mean of cellwise ratios (1) is the largest of the five at every size, reaching 17.6 at the largest size where the pooled estimand reads 10.9. The reason is mechanical: a per-level ratio at a level with a small reference score norm can be very large while contributing little energy, and averaging ratios weights all levels equally. Pooling the energies first weights each level by how much score there is to get wrong, which is exactly what the pooled definition specifies. Earlier drafts reported (1) as the headline; the range 7.0–17.6 they quoted is therefore an artefact of that estimand, and the pooled range 6.6–10.9 replaces it. The ratio of per-seed averaged errors (2) runs 6.6 to 12.4.

No inequality orders these summaries in general: Jensen gives $\mathbb{E}[1/Y] \geq 1/\mathbb{E}[Y]$, but $\mathbb{E}[X/Y] \geq \mathbb{E}[X]/\mathbb{E}[Y]$ does not follow for arbitrary positive correlated X and Y. The observed ordering is a property of these cells, which is exactly why the estimand has to be named rather than assumed.

Resolving the average over noise levels shows where any single number hides structure: the per-level ratio is not flat in t but peaks near $t \approx 0.5$ and falls at both ends. Two different "weakest" quantities have to be kept apart. The weakest seed-averaged (M, t) aggregate in the grid is 3.6; the weakest individual cell is 0.92, and EM–BP is beaten outright in 2 of the $1,520$ cells summarised here. The estimator's advantage is largest at moderate noise, where the posterior is dominated by neither the likelihood nor the prior — which is where knowing the transition structure should help most.

Table 7. the archival structured comparison (Appendix G)'s all-site ratio under four aggregations of the same cells, with the paired bootstrap interval over seeds. (1) is what the table reports. The conclusion does not depend on the choice: no estimand at any size falls below 1.98×, and the largest departure from the reported value is 0.48.

M	(1) per-seed	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (1)
32	2.34	2.00	2.15	1.98	[1.88, 2.86]
128	2.82	2.45	2.61	2.62	[2.29, 3.44]
512	3.58	3.43	3.51	3.59	[3.25, 3.91]
2048	6.14	5.66	5.89	5.93	[5.30, 7.02]